Upper GI bleeding (non-variceal)

Risk stratification

- **Glasgow–Blatchford** (pre-endoscopy; 0–1 → outpatient), Rockall (post-endoscopy)
- Forrest classification of ulcer stigmata (Ia spurting → III clean base)

Resuscitation

- Restrictive transfusion: Hb 7 g/dL (8 if cardiac disease)
- IV PPI, erythromycin before endoscopy; correct coagulopathy; stop NSAIDs/aspirin review

Endoscopy and beyond

- Endoscopy within 24 h (urgent if unstable)
- Dual therapy (adrenaline + clip/thermal) for Forrest Ia–IIb
- Rebleed: repeat endoscopy → **interventional radiology embolisation** → surgery
- TXA not beneficial (HALT-IT)

Key point: Transfuse restrictively; endoscope within 24 h.

Variceal bleeding

Before endoscopy

- Restrictive transfusion (Hb 7–8), avoid overfilling
- **Terlipressin** (or octreotide), antibiotic prophylaxis (ceftriaxone 1 g/day)
- Airway protection if encephalopathy/massive haematemesis

Endoscopic and rescue therapy

- **Band ligation** (oesophageal); cyanoacrylate for gastric varices
- Balloon tamponade (Sengstaken) or self-expanding stent as bridge
- **Pre-emptive TIPS within 72 h** in Child C 10–13 or B with active bleeding

Rebleeding prevention

- Non-selective beta-blocker (carvedilol) + repeat banding

Key point: Antibiotics + vasoactive drug + banding = standard bundle.

Notes: GIT full topics p.10

Paracetamol-induced acute liver failure

Definition

- Coagulopathy (INR > 1.5) + encephalopathy within 26 weeks, no prior liver disease
- O'Grady: hyperacute (< 7 d), acute (8–28 d), subacute (> 28 d)

King's College criteria (paracetamol)

- pH < 7.3 after resuscitation, or all of: INR > 6.5, creatinine > 300 µmol/L, grade 3–4 HE
- Lactate > 3.5 (4 h) or > 3 (12 h) added

ICU management

- **NAC regardless of time**; neuroprotection (Na 145–150, head up), glucose
- ICP concerns: high ammonia > 150; RRT early (lowers ammonia)
- Avoid FFP unless bleeding; refer early to transplant centre

Key point: Refer early: deterioration is rapid.

Hepatic encephalopathy

Classification

- West Haven grades 1–4; type A (ALF), B (bypass), C (cirrhosis)
- Precipitants: infection, GI bleed, constipation, electrolytes, sedatives, dehydration

Management

- Treat precipitant; **lactulose** to 2–3 soft stools/day; **rifaximin** for recurrence
- Airway protection if grade 3–4

Ammonia

- High ammonia correlates with severity in ALF (> 150–200 → cerebral oedema)
- In cirrhosis, levels not useful for diagnosis/monitoring

Key point: Look for the trigger in every episode of HE.

Notes: GIT full topics p.13

Acute-on-chronic liver failure

Definition

- Acute decompensation of cirrhosis + organ failures + high short-term mortality
- **CLIF-C ACLF** grades 1–3

Organ support

- Albumin for SBP/HRS, terlipressin; vasopressors (MAP ≥ 65)
- Restrictive transfusion; avoid nephrotoxins; nutrition
- Treat infection early; consider steroids for alcoholic hepatitis

Prognosis

- ACLF-3: 28-day mortality > 70%
- Reassess at day 3–7; transplant evaluation; palliative care

Key point: ACLF grade guides the ceiling of care discussion.

Notes: GIT full topics p.26

Spontaneous bacterial peritonitis

Interpretation

- **Ascitic neutrophils > 250/mm³** = SBP; culture often negative
- SAAG ≥ 11 g/L = portal hypertension

Management

- **Cefotaxime/ceftriaxone** (carbapenem if nosocomial)
- **Albumin 1.5 g/kg day 1, 1 g/kg day 3** ↓ HRS and mortality
- Prophylaxis: norfloxacin/ciprofloxacin after SBP

Large-volume paracentesis

- Albumin 8 g/L removed when > 5 L (prevents paracentesis-induced circulatory dysfunction)

Key point: Albumin with antibiotics in SBP reduces kidney injury.

Acute pancreatitis: severity and early care

Revised Atlanta

- Mild: no organ failure; moderate: transient OF < 48 h or local complications
- Severe: persistent OF > 48 h (modified Marshall)

Early management

- **Moderate goal-directed fluid** (WATERFALL: aggressive harms)
- Analgesia (opioids OK), early enteral nutrition within 24–48 h
- No prophylactic antibiotics; ERCP within 24 h only for cholangitis

Complications

- Systemic: ARDS, AKI, shock, ACS, hypocalcaemia
- Local: APFC, pseudocyst, ANC, WON, infected necrosis

Key point: Organ failure duration defines severity.

Infected pancreatic necrosis

Diagnosis

- Clinical deterioration after week 2; gas on CT; FNA rarely needed

Management

- Carbapenem/quinolone + metronidazole
- **Step-up approach**: percutaneous or endoscopic drainage → necrosectomy (PANTER, TENSION)
- Delay intervention to ≥ 4 weeks (walled-off) (POINTER)

Early surgery

- Abdominal compartment syndrome, bowel ischaemia, perforation, haemorrhage

Key point: Step up and delay: minimally invasive first.

ICU nutrition

Timing and dose

- **Early EN within 48 h** if haemodynamically stable (low dose on vasopressors)
- Energy 70% of target first week (indirect calorimetry), protein 1.3 g/kg/day progressively
- EFFORT Protein: high protein no benefit (harm in AKI)

Feed intolerance

- GRV > 500 mL: prokinetics (metoclopramide, erythromycin), post-pyloric
- Assess abdomen, constipation, ischaemia

Parenteral nutrition

- If EN contraindicated/fails after 3–7 days; early PN if high risk/malnourished
- EPaNIC: late PN better

Key point: Start enteral feeding early but slow; avoid overfeeding.

Notes: GIT full topics p.25

Refeeding syndrome

Pathophysiology

- Insulin surge → intracellular shift of phosphate, K, Mg; thiamine depletion
- Results: heart failure, arrhythmia, respiratory failure, Wernicke's

High-risk (NICE)

- BMI < 16, weight loss > 15%, little intake > 10 days, low K/P/Mg before feeding

Management

- **Start 5–10 kcal/kg/day**, increase over 4–7 days
- Thiamine before feeding; replace P, K, Mg; monitor daily

Key point: Feed slowly and replace phosphate in high-risk patients.

Toxic megacolon and acute severe colitis

Definition and causes

- Colonic dilatation > 6 cm + systemic toxicity
- Causes: UC, Crohn's, C. difficile, CMV, ischaemia

Medical management

- **IV hydrocortisone 100 mg 6-hourly**; stop opioids/anticholinergics
- Fluids, K, VTE prophylaxis, rule out C. difficile/CMV
- Rescue: infliximab or ciclosporin at day 3–5 if no response

Surgery

- Perforation, haemorrhage, worsening, no response to rescue → colectomy

Key point: Day 3 review: no response → rescue therapy or surgery.

Intra-abdominal hypertension

Definitions (WSACS)

- IAH ≥ 12 mmHg (grades I–IV); ACS > 20 + new organ dysfunction
- APP = MAP − IAP (target ≥ 60)

Organ effects

- ↓ Venous return/CO, ↑ airway pressures, AKI (renal compression), gut ischaemia, ↑ ICP

Management

- Evacuate gut (NGT, rectal tube), drain ascites, sedation/NMB
- Negative balance, avoid tight dressings; **decompressive laparotomy** if ACS

Key point: Measure bladder pressure in at-risk patients.

Acute mesenteric ischaemia

Types

- Arterial embolism (AF), arterial thrombosis, venous thrombosis, non-occlusive (NOMI)

Diagnosis

- Pain out of proportion; lactate late; **CT angiography**

Management

- Fluids, anticoagulation, antibiotics
- Revascularisation (endovascular/surgical) + resection of dead bowel; second look
- NOMI: treat low flow, vasodilators (papaverine)

Key point: Pain out of proportion in AF → CT angiography.

ALF complications and liver support

Cerebral oedema

- Ammonia → glutamine → astrocyte swelling, ↑ ICP; herniation

Emergency management

- Head up, Na 145–155, hyperventilation briefly, mannitol/HTS
- RRT to lower ammonia; temperature 35–36

Liver support and transplant

- Plasma exchange (high-volume: Larsen 2016 benefit)
- MARS not proven; **emergency transplantation** (King's criteria)

Key point: Ammonia > 150 → protect the brain.